

Decision Tree:

The main goal : Find the root node

1) Information gain: Entropy

Step – 1: Calculate Individual entropy

Step – 2: Calculate total entropy

Step – 3: Calculate gain = $H_{Total} - H_{feature} = step_2 - step_1$

Step – 4: Split information of each feature

Step – 5: Gain ratio = $\frac{Gain}{Split\ information} = \frac{step3}{step4}$

2) Gini index:

- Information gain will talk about positive way: Clarity in information
- *Gini index will give the idea about impurity, how much a feature has impure*
- Impure means : How a data sample is misclassified
- For example, assume that a person does not have cancer
 - If model says as cancer : wrong prediction
- This is misclassified, It is impure
- If a data has one class : Perfectly Pure
- If a data has multiple classes: Impure

The behaviour of root node is less impurity, Gini index should be low

$$Gini = 1 - \sum_{i=1}^n p_i^2$$

NOTE: Gini uses at a Time only Binary split

Example:

<i>Color</i>	<i>Class</i>
<i>RED</i>	<i>A</i>
<i>RED</i>	<i>A</i>
<i>Blue</i>	<i>B</i>
<i>Blue</i>	<i>B</i>

*here RED belongs to class A and Blue belongs to class B ,Its Pure
if Red and Blue belongs to A ,it mixed split .Its Impure*

For example income feature:

<i>Id</i>	<i>income</i>	<i>Buys a computer</i>
1	High	No
2	High	No
3	High	Yes
4	Medium	Yes
5	Low	Yes
6	Low	No
7	Low	Yes
8	Medium	No
9	Low	Yes
10	Medium	Yes
11	Medium	Yes
12	Medium	Yes
13	High	Yes
14	Medium	No

Income divide into 3 splits

- *High*
- *Low*
- *Medium*

{ }, {High}, {Low}, {Medium}, {High, Low}, {Low, Medium}, {High, Medium},

$\{High, Medium, Low\}$: Power set or Super set

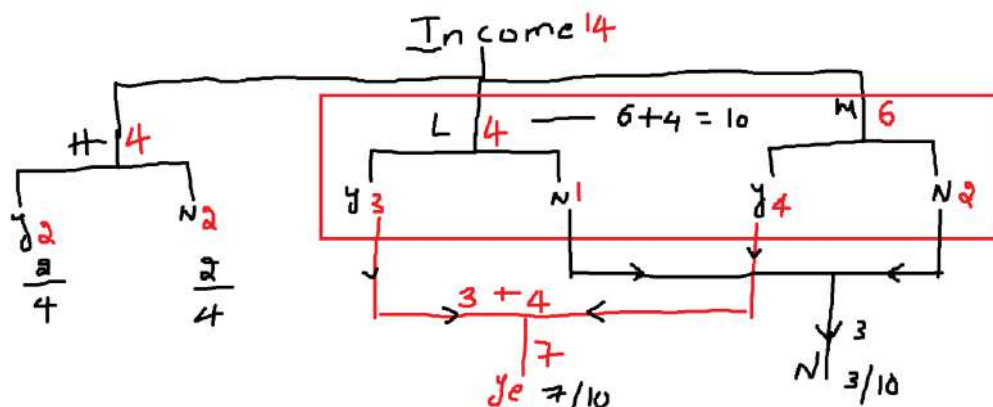
Goal: Split pure or Impure binary split for fast compute

Under the Gini of income:

Case – 1: $\{High, low\}$ and $\{Medium\}$

Case – 2: $\{High, medium\}$ and $\{low\}$

Case – 3: $\{low, medium\}$ and $\{high\}$



Step – 1: Calculate total data Gini (Before split)

Yes = 9 ; No = 5

$$p(\text{yes}) = \frac{9}{14} \quad p(\text{No}) = \frac{5}{14}$$

$$Gini_{data} = 1 - \sum_{i=1}^{n=2} p_i^2 = 1 - [p_{yes}^2 + p_{no}^2]$$

$$1 - [p_{yes}^2 + p_{no}^2] = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2$$

Step – 2: Calculate total data Gini index of income

Case – 1: $\{Low, medium\}$ vs $\{High\}$

$$Gini_{income} = w_{lm} * Gini_{lowmedium} + w_{high} * Gini_{high}$$

$Gini_{lowmedium}$:

$$p(\text{yes}) = \frac{7}{10} \quad p(\text{No}) = \frac{3}{10}$$

$$1 - [p_{yes}^2 + p_{no}^2] = 1 - \left(\frac{7}{10}\right)^2 - \left(\frac{3}{10}\right)^2$$

$$w_{lm} = \frac{10}{14}$$

Gini_{high}:

$$p(\text{yes}) = \frac{2}{4} \quad p(\text{No}) = \frac{2}{4}$$

$$1 - [p_{\text{yes}}^2 + p_{\text{no}}^2] = 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2$$

$$w_{\text{high}} = \frac{4}{14}$$

$$Gini_{\text{income}} = w_{lm} * Gini_{\text{lowmedium}} + w_{\text{high}} * Gini_{\text{high}}$$

$$Gini_{\text{income}_{\text{case1}}} = \frac{10}{14} * \left[1 - \left(\frac{7}{10}\right)^2 - \left(\frac{3}{10}\right)^2\right] + \frac{4}{14} * \left[1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2\right] = 0.443$$

similarly we need to calculate

case – 2: {High medium} vs {Low}

$$Gini_{\text{income}_{\text{case2}}} = w_{hm} * Gini_{\text{highmedium}} + w_{\text{low}} * Gini_{\text{low}} = 0.45$$

$$Gini_{\text{income}_{\text{case2}}} = \frac{10}{14} * \left[1 - \left(\frac{6}{10}\right)^2 - \left(\frac{4}{10}\right)^2\right] + \frac{4}{14} * \left[1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2\right] = 0.45$$

case – 3: {High low} vs {medium}

$$Gini_{\text{income}_{\text{case3}}} = w_{hl} * Gini_{\text{higlowlow}} + w_{\text{med}} * Gini_{\text{med}} = 0.458$$

$$Gini_{\text{income}_{\text{case3}}} = \frac{8}{14} * \left[1 - \left(\frac{5}{8}\right)^2 - \left(\frac{3}{8}\right)^2\right] + \frac{6}{14} * \left[1 - \left(\frac{4}{6}\right)^2 - \left(\frac{2}{6}\right)^2\right] = 0.458$$

$$Gini_{\text{income}_{\text{case1}}} = 0.443$$

$$Gini_{\text{income}_{\text{case2}}} = 0.450$$

$$Gini_{income_{cases}} = 0.458$$

Gini varies from 0 to 1

- 0 : pure
- 1: highly impure

Conclusion is: Case1 is the best split: {Low, medium} vs {High}

we have total 4 input columns

- *age*
 - *Young*
 - *MA*
 - *senior* ===== 3 cases out of 3 cases, 1 is the best split
- *Income*
 - *High*
 - *Medium*
 - *Low* ===== 3 cases out of 3 cases, 1 is the best split
- *Student*
 - *Yes*
 - *No* ===== 1 case
- *Credit rating*
 - *Fair*
 - *Excellent* ===== 1 case

*For first split we need to calculate 10 cases Gini, out of 10 we need to select only one split best split
this process will continue till reach leaf level*

Step – 3: Change in Gini

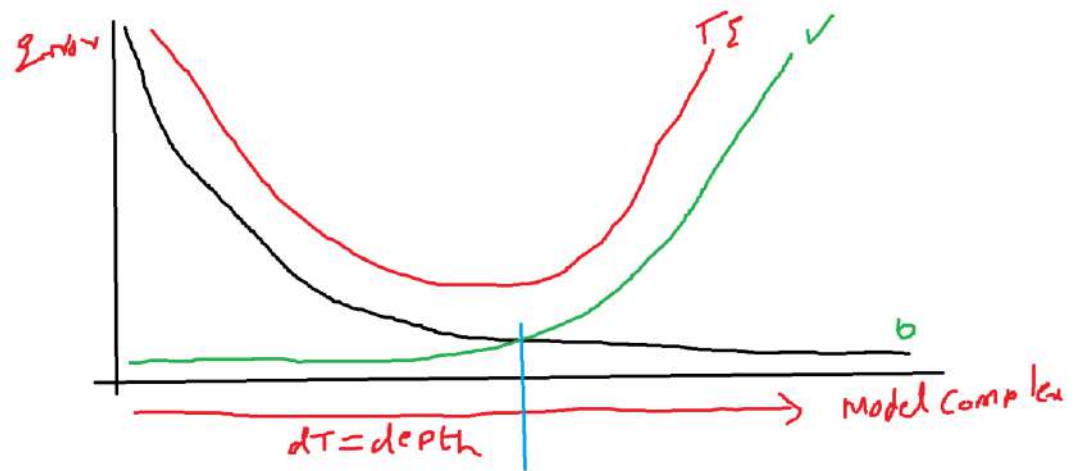
$$\Delta Gini_{feature} = Gini_{total} - G_{feature}$$

Conclusion: The change in Gini should be high to get a Root node

it indicates the corresponding feature reducing maximum impurity

Gini also called as CART algorithm : Classification and Regression Tree

Tree Pruning



As Tree Depth increases, The model Complexity increases

Less Depth = Under fit

More Depth = Overfit

Tree Pruning Two types

- **Pre pruning**
 - Analogy with Regression equation
 - Every time will add a column will check the R_{Sq} and Adj_R
 - Forward selection method
 - Similarly we need to increase the depth of the tree level by level
 - at each level will calculate the Performance of the Model
- **Post Pruning**
 - Analogy with Regression equation
 - Add all column will check the R_{Sq} and Adj_R by removing each variable
 - Backward selection method
 - Similarly let the Tree grown first as much as it can
 - then decrease the Depth level by level